Referee report on `mjl_part4_draft2.tex`

"A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, IV", Draft 2

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0. Scope

Read in full. Draft 2 (966 lines). It compiles to 15 pages without errors.

Checked against the attached sources. Every statement that Draft 2 cites from:

- Part I, Draft 6;
- Part II, Draft 6 (theorem numbers only);
- Part III, Draft 6;

FK (§§3, 4, 6, and the Introduction, for $(n, N) = 1$);

- Zou, arXiv v3: (2.2), (3.2), Remark 3.1, Lemma 3.5, and the definition of x^* on p. 8.

For Parts I–III I relied on my reading for the Draft 1 report and rechecked only the cited items.

Version caveat. Draft 2 says it was checked against Part I, Draft 5, Part II, Draft 4, and Part III, Draft 7. I have Part III only in Draft 6. All the Part III numbers Draft 2 cites agree with Draft 6:

- Lemma 3.3;
- Theorems 4.3, 7.6 and 9.1;
- Corollaries 5.1 and 5.2;
- Lemma 7.5;
- Proposition 11.3.

If Draft 7 renumbered anything, these must be rechecked.

Numbering used below (compiled).

- Proposition 2.1 (diagonal commutator);
- Theorem 3.1 (FK good elements);
- Example 3.2 and Remark 3.3;
- Lemma 4.1 and Proposition 4.2 (kernel, index, modulus);

- Remark 4.3 and Example 4.4;
- Proposition 5.1 (Hasse principle) and Theorem 5.2 (global good classes);
- Proposition 6.1 (lattice criterion);
- Proposition 7.1 (Zou constant);
- Proposition 8.1 (conditional comparison);
- Proposition 9.1.

Summary verdict

Draft 2 is a large improvement. It deals with every substantive point of my Draft 1 report:

- the Flicker–Kazhdan good-element theorem is attributed and labeled **[E]**;
- the global good-class criterion is now Theorem 5.2;
- the two global comparisons are clearly separated;
- b_D is shown to be independent of the choice of measures;
- $(n, N) = 1$ and Zou are cited;
- the central characters are fixed in the right order.

I found no false mathematical statement. One proof step is wrong as stated: the general Haar-modulus claim in Proposition 4.2. Its conclusion is right, but the stated reason is false in general (§1.3).

The main remaining issues are:

(1) Proposition 8.1 is close to a tautology. It should be replaced by a precise local matching condition at split places, which [FK, §6] already supplies (§1.6).

(2) The paragraph on Li's trace formula says something false about real places and makes an unsupported claim (§1.7).

(3) Drafting history is still in the body (§3).

Result	Mark	Comment
Prop. 2.1	correct	Rechecked; the c/c_v issue is resolved
§2.2 (star map)	correct; minor repair	Say why $u(x) = 1$
Thm. 3.1 [**E**]	correct	Proof verified; attributed
Ex. 3.2, Rem. 3.3	correct	The regularity remark is valuable
Lemma 4.1	correct	
Prop. 4.2	correct; proof repair	The general "$[A : I]/|K|$" rule is false; the formula holds for another reason
Prop. 5.1 [**E**], Thm. 5.2	correct	
§§6.1, 6.3 (example)	correct	I checked $\Delta(t)/\Delta(t^3)$ and the values 9 and 27
Prop. 6.1	correct	
Prop. 7.1	correct; minor repair	"Provided x^n is regular" is automatic; justify the central-character hypothesis
Prop. 8.1 [**C**]	correct but nearly empty	Replace (i)–(ii) by the FK §6 matching condition
§8.4 (Li)	false statement	K has no real places; "known" is uncited
Prop. 9.1	correct	This is Prop. 6.1 again; consider merging

1. Correctness

1.1 Section 2

Proposition 2.1. Correct. It is the computation I verified for Draft 1. The distinction between c and c_v is now explicit, and the remark on $(r, n) = 1$ is right.

Star map (§2.2). Correct, with one justification to add.

- *The power identity.* The identity $(\tilde{h}_1\tilde{h}_2)^n = \tilde{h}_1^n\tilde{h}_2^n[\tilde{h}_2, \tilde{h}_1]^{n(n-1)/2}$ for a central commutator, and its vanishing for n odd, are fine.
- *Why $\zeta^* = 1$.* The products $\sigma(\zeta^k I, \zeta I)$ are powers of $(\zeta, \zeta)_n = (\zeta, -1)_n = 1$. I checked this.
- *Why $u(x) = 1$.* "For odd n the correction $u(x)$ is identically 1" is true, but Zou's Lemma 3.5 proves only that the *ratio* $u(x)/u(x_1)u(x_2)$ is 1 for odd n. The real reason is on Zou, p. 8: $u(x) = \pm 1$ "as in [FK, §4]", and $u(x)$ lies in μ_n. For odd n, $\{\pm 1\} \cap \mu_n = \{1\}$. Say this.

1.2 Section 3

Theorem 3.1. Correct. I checked the proof:

- the projection formula $(a, N_{E_i/F}b)_F = (a, b)_{E_i}$ for $a \in F^\times$;
- the step $a^{r(1+2c_v)} = a \cdot a^{r-1+2rc_v}$.

Remark 3.3. Correct and useful. Two small additions:

- In the example, $\mathrm{diag}(1, \zeta_3)$ is regular but not good, since $\zeta_3 \notin F^{\times 3}$ (because $\zeta_9 \notin \mathbb{Q}_3(\zeta_3)$).
- The last sentence (field tori preserve regularity) has a one-line proof that should be included. If $t^n \in F$, then $[F(t) : F]$ divides both n and r.

1.3 Section 4

Lemma 4.1. Correct.

Proposition 4.2, kernel and index. Correct. I checked:

- $\ker P_n \cong \mu_n^k/\mu_n$, using $\mu_n(E_i) = \mu_n(F)$;
- $|M^\times/M^{\times n}| = n^2|n|_M^{-1}$;
- the quotient computation, using Lemma 4.1;
- Example 4.4: the values $1, 9, 9$ and $3, 81, 27$.

Proposition 4.2, the Haar formula: proof repair. The formula

$$\int_{A_E} q(a^n)\,da = n^{k-1}|n|_F^{-(r-1)} \int_{P_n(A_E)} q(b)\,db$$

is correct. The stated reason is not. The proof, and Remark 4.3, assert that for an open endomorphism φ with finite kernel K and image I of finite index, the density is $[A : I]/|K|$. This fails already for $A = \mathbb{Z}$ (counting measure) and $\varphi(a) = na$. The pushforward of counting measure is counting measure on $n\mathbb{Z}$, so the density is 1, not n.

The correct general statement: take a compact open subgroup U with $U \cap K = 1$. Then $\varphi^{-1}(\varphi(U)) = UK$, and

$$\lambda = \frac{\mathrm{vol}(UK)}{\mathrm{vol}(\varphi U)} = |K|\,[U : \varphi(U)].$$

For A_E, take U a small torsion-free open subgroup of $\mathcal{O}_E^\times/\mathcal{O}_F^\times$. This is a $\mathbb{Z}_p$-module of rank $(r-1)[F : \mathbb{Q}_p]$, so $[U : U^n] = |n|_F^{-(r-1)}$, and $\lambda = n^{k-1}|n|_F^{-(r-1)}$.

That this equals $[A_E : I]/|K|$ is a coincidence of this group: the valuation lattice has rank $k-1$ and $|K| = n^{k-1}$. Remark 4.3 should therefore read: "the factor is $|\ker P_n|$ times the local index $[U : U^n]$, not the index $[A_E : P_n A_E]$".

1.4 Section 5

Proposition 5.1 and Theorem 5.2. Correct. They agree with the corollary I proposed, and the proof is complete. Two remarks:

- The D-side case $v \notin S$ goes through Part I, Proposition 2.6 (the isomorphism $\overline{G_D}(K_v) \cong \overline{G}_v$). Cite it by number, not as "the exact Part I cover isomorphism".
- Remark 5.3 reproduces Part I's $c(\gamma)$ exactly (Part I, §6.1). Its last sentence is an important point, made correctly.

1.5 Sections 6 and 7

§6.1. The formula for L_D agrees with Part III, Definition 4.1. The formula for Δ is correct: the eigenvalues of $\mathrm{Ad}(t)$ are λ_i/λ_j. The statement about the identity value is correct.

§6.3, the explicit example. Correct. I checked:

- $\Delta(t)/\Delta(t^3) = |\lambda_1\lambda_2|/|\lambda_1^2 + \lambda_1\lambda_2 + \lambda_2^2|$;
- $\lambda_1^2 + \lambda_1\lambda_2 + \lambda_2^2 = \mathrm{Trd}^2 - \mathrm{Nrd}$;
- the limit 9 at scalars, giving the value 27.

The paragraph on twisted finite quotients corrects my own Draft 1 suggestion, which was sloppy: $\tilde{H}/\tilde{Z}U$ loses the genuine central character.

Proposition 6.1. Correct, with the redundancy removed.

Proposition 7.1. Correct; I checked $1/(d_F(n)|n|_F^{r/2}) = b_D/n$. Two repairs:

- *A redundant proviso.* "Provided x^n is regular" is automatic for a field centralizer, by Remark 3.3. State it as a consequence.
- *The central-character hypothesis.* Zou, Remark 3.1, says Zou deliberately removes the central-character restriction. So the hypothesis $\omega_\pi|_{\mu_n} = 1$ is a genuine restriction, not a convention. Note what happens without it:
  - if $\omega_\pi|_{\mu_n} \neq 1$, the root sum $\sum_\zeta \omega_\pi(\zeta)$ vanishes, so (3.2) forces $\theta_{\tilde{\pi}}(x^*) = 0$ at elliptic x;
  - for a lift pair, the compatibility $\omega_\pi(a) = \omega_{\tilde{\pi}}(a^*)$, together with $\zeta^* = 1$, gives $\omega_\pi|_{\mu_n} = 1$ automatically.

  Saying this would make the proposition self-contained.

1.6 Section 8: replace Proposition 8.1

Proposition 8.1 is correct but carries little content. Hypothesis (ii) is essentially the conclusion. More importantly, it leaves the split-place question vague ("after any prescribed local root summations"). The question has a definite answer.

Why a root sum cannot be the local identity at split places. The $D^\times$-side sum runs over rational classes $[s]$. The cover side runs over $[s^n]$, in bijection with them by Theorem 5.2. At $v \notin S$, the local roots ζs, with $\zeta \in \ker P_n$ (Proposition 4.2), are not rational. So a local identity of the root-sum form

$$J(s^n, \tilde{f}_v) = \lambda_v \sum_\zeta J(\zeta s, f_v)$$

cannot be used term by term. What is needed is

$$J(s^n, \tilde{f}_v) = \lambda_v \, J(s, f_v) \quad \text{for all regular } s.$$

This forces $J(\cdot, f_v)$ to be constant on cosets of $\ker P_n$.

FK already impose exactly this. [FK, §6] state that they "deal only with functions f with the property $\omega(z)F(x,f) = F(x',f)$ for any x, x' with $zx^* = x'^*$". (I read this in the attached scan.) Proposition 8.1(i)–(ii) should be replaced by:

(a) at each $v \notin S$, the ordinary test function satisfies the FK §6 invariance;

(b) a pointwise local identity holds with an explicit constant λ_v;

(c) the product of all the λ_v, the values $b_{D,v}$ at $v \in S$, and the archimedean factors equals 1.

A second constraint, from Remark 3.3. Take s regular with s^n singular (e.g. $s = \mathrm{diag}(1, \zeta)$). The map $s \mapsto s^n$ is a local homeomorphism of the torus. So $J(s^n, \tilde{f}_v)$, pulled back to a neighborhood of s, must be locally constant there, because $J(\cdot, f_v)$ is locally constant on the regular set. But near a central s^n, $J(\cdot, \tilde{f}_v)$ has a non-constant germ expansion. So the existence of the matching f_v imposes vanishing conditions on germs of $\tilde{f}_v$. Alternatively it forces a support condition, as in Zou's restriction to x with x^n regular.

This should be written into the ledger. It is a precise, checkable obligation, and it shows that matching at split places is not merely a pullback.

The rest of §8. §8.1–§8.2 are correct:

- the product formula $\prod_v \Delta_v(\gamma) = 1$;
- the local identity at $v \in S$ (Part III, Theorem 9.1);
- the formal-degree conversion $d(\mathrm{St})\, dg_{D,v}$.

The citation "Part II, Theorem 6.2" is right: `thm:comparison2` is 6.2, and Theorem C′ is 6.5. Delete the phrase "in the numbering cited by the referee".

1.7 §8.4: Li's trace formula

"Under the trace Paley–Wiener hypothesis at real archimedean places ... This hypothesis is known for the cited metaplectic covers of GL" needs correcting:

- K is totally imaginary, so there are no real places.
- Li's hypothesis [Li4] concerns the invariant Paley–Wiener theorem at the archimedean places.
- At complex places the covers split (Part I), so the hypothesis reduces to the linear case. That case is a known theorem, but it needs a citation (e.g. Clozel–Delorme), which the draft does not give.
- "Known for the cited metaplectic covers" is unsupported as it stands.

Part II, §7(iii), has the right wording.

1.8 Section 9

§9.1. Correct.

- Note that $\theta(a) = \tilde{\theta}(a^*)$ is automatically trivial on $K^\times$, since $a^* = \iota(a)^n = \iota(a^n)$ and $\tilde{\theta}$ is trivial on $\iota(K^\times)$. Worth one sentence.
- Proposition 9.1 restates Proposition 6.1. Its **[C]** label is misleading: the implication itself is proved. Merge it into Proposition 6.1 as a remark.

2. Consistency and citations

(1) *Labels.* **[R]** is gone. The meanings of **[O]** and **[C]** still differ from Part I's ("proof outlined", "conjectural"). This needs a series-wide decision.

(2) *Titles.* The title now follows Parts I–II. But the bibliography entry [JoynerIII] still gives Part III's old "Toward . . . " title. The author line (Joyner, ChatGPT, Claude) differs from Parts I–II. "D. Joyner with collaborators" in the bibliography should give the actual authors.

(3) *Citation style.* Only DKV, Kottwitz, Li and Zou are cited with `\cite`. FK, KP and Parts I–III are typed in by hand. [Takeda] is in the bibliography but never cited.

(4) *Unverified.* Not checked by me:

- Takeda's page range: Draft 2 gives 415–449, Draft 1 gave none; I could not check it;
- KP (Publ. Math. IHÉS 59, 35–142), which agrees with the standard citation as far as I know;
- the DKV page range;
- Kottwitz and Li, which I did not have.

(5) *Checked.* The FK and Zou items listed in §0; Part I's Proposition 2.3(b), Proposition 3.1, Proposition 3.5(a), §6.1 and $c(\gamma)$; Part II's Definition 2.1 and theorem numbers.

3. Readability

(R1) *Drafting history is still in the body.* Move these to the log:

- the paragraph "Scope of direct cross-checking" (§1), which lists drafts and mentions "Claude's attached referee report";
- in §2.1, "this is exactly the cocycle used in Part IV, Draft 1; its letter c meant . . . ";
- in §8, "in the numbering cited by the referee".

The acknowledgments may credit the referee report, as Parts I–II do. "Suggested repairs to the chronology of the series" is unclear and should be reworded or cut.

(R2) *Disclaimers.* These are reduced, but they still recur:

- the end of the abstract;
- the status convention ("No representation-theoretic correspondence follows . . . ");
- Remark 7.2;
- §8 ("not a formal consequence");
- the last paragraph of §9;
- the acknowledgments.

The status table suffices.

(R3) *Orienting paragraphs.* These are good throughout.

(R4) *Glossary.* It is now short and specific to the paper. That is acceptable.

(R5) *Abstract.* It is accurate. "Shows why regularity of a root must be distinguished from regularity of its nth power" refers to Remark 3.3, not to Proposition 4.2. Reword.

4. Recommendation

Minor revision for §§2–7, after these changes:

- repair the proof of Proposition 4.2 and Remark 4.3;
- justify $u(x) = 1$;
- make the hypothesis of Proposition 7.1 explicit.

For §8, the most valuable next step is to replace Proposition 8.1 by the FK §6 matching condition at split places, together with the germ constraint of §1.6. This would turn §8 into a precise statement of the remaining local theorem. Better still, the theorem could be proved:

- for $v \nmid n$, via FK's unit-element and matching results under $(n, N) = 1$;
- at $v \in S$, using Part III, Theorem 9.1.

The explicit wild test of §6.3 remains the cleanest way to settle the first case of the integrality question. It can be done as a computation in a twisted group algebra in GAP or Sage.